

Introduction on Laboratory Tests for Diagnosis of Infectious Diseases and Immunological Disorders

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Introduction

Laboratory tests are extremely valuable in validating a diagnosis, predicting disease severity, and also in following up on the disease activity in patients who are suspected of having infectious diseases or immunological disorders. Diagnostic reliability and health benefits could be considered as two important factors when testing for such diseases. Meanwhile, timely diagnostic assessment and implementation of accurate tests are highly valuable in disease management (Caliendo, 2013). Microbiological methods, including culture and staining, have previously become the gold standard for detecting infectious diseases. These techniques can directly detect microorganisms such as bacterial, viral, parasitic, or fungal agents in fluids or tissues of the patient. Although these tests are reliable, characteristics such as time-consuming testing hamper quick diagnosis (Laupland and Valiquette, 2013). Currently, utilizing indirect laboratory techniques to evaluate infectious diseases is of utmost importance for detecting specific antibodies or pathogen antigens in blood or body fluids. Several laboratory methods have been designed for detecting either pathogens or specific antibodies in the suspected subjects. Pathogen isolation and serological methods are the conventional diagnostic tests that are widely applied to recognize the cause of a disease (Amar, 2014). Furthermore, molecular diagnostics became an important tool in healthcare in the early 1990s, when specific monoclonal antibody technology was developed. Molecular tests have higher sensitivity

and specificity compared to traditional methods. These methods can be helpful in the surveillance and control of infectious diseases, but more precise and sensitive diagnostic methods are still required (Lu et al., 2020). In this article, we will review the most important laboratory tests that are widely used to diagnose infectious diseases or various immunological disorders, including autoimmune and immunodeficiency disorders.

Diagnostic laboratory tests: Focusing on infectious diseases

Direct detection of microorganisms: Culture, staining, microscopy

Culture is usually the gold standard for identifying organisms, but results can take days or weeks. Furthermore, because culture cannot be performed for all infections, alternative tests are required. Following culture and identifying the organism, evaluation of the pathogen's susceptibility to antimicrobial drugs can also be performed. This can be helpful in implementing the appropriate antimicrobial therapy. Molecular methods can also be used to find particular resistance genes.

In order to identify a pathogen and microscopic detection, different staining methods can be used. Gram staining (using crystal violet dye) is frequently used in microbiology laboratories to classify bacteria into gram-positive or gram-negative and to characterize bacteria morphology (e.g., cocci, bacilli) (Srinivasan et al., 2012). Another staining method known as acid-fast staining (or Ziehl-Neelsen staining) is used to identify mycobacterium species. Carbol fuchsin is used in acid-fast staining that can dissolve the lipid components of the Mycobacteria cell wall. Using heat allows further penetration and therefore dye entrance into the cytoplasm. As a result, all cells appear red. Acid fast cells remain red following the decolorization step, while non-acid fast organisms become colorless. So, the colorless cells appear blue following staining with methylene blue in the next step. Therefore, the red and blue colors represent acid-fast and non-acid fast bacteria, respectively (Lamanna, 1946).

Serological methods: Assay based on antigen-antibody reaction

"Max von Gruber" was the first to use serologic methods to diagnose infectious agents (Von Gruber and Durham, 1896). He used sera from infected animals and humans to agglutinate typhus and cholera bacteria. The invention of the complement fixation procedure by "Bordet and Gengou" in 1901 is also regarded as another milestone in detecting infectious agents through serologic methods (Silverstein, 2009). Serologic tests are simple and inexpensive methods that provide immediate results. Serologic tests for infectious diseases can provide information about the disease even when there is no virus or its components in the sample. In addition, these tests can also confirm the presence of infection in asymptotically infectious diseases (Vainionpää et al., 2015).

In general, serologic tests are based on the specific interaction between antigen and antibody. Antigens can be a part of a pathogen like bacteria and viruses or a non-pathogenic agent such as red blood cells (RBC), hormones, tissue proteins, etc. Binding of antigen to an antibody is a specific reaction that can be influenced by several factors such as the pH of the environment (optimum pH is approximately 7.2), ionic strength, concentration ratio of antigens and antibodies, environment's temperature, and incubation time (Vainionpää et al., 2015). Optimal reaction between antigen and antibody and forming antigen-antibody complexes can occur in the appropriate condition. Importantly, the higher or lower concentration of either antibody or antigen causes false negative results. Higher concentrations of antibodies, for example, in a variety of tests such as Wright and Widal (serologic tests for the detection of Brucellosis and typhoid, respectively), can result in a phenomenon known as the "Prozone effect," in which the ability of antibodies to form immune complexes is impaired. Similarly, the excess concentration of antigens in tests such as CRP-latex (for detection of C-reactive protein in the serum) can result in the "Postzone effect" causing inappropriate formation of antigen-antibody complexes (Jin and Zehnder, 2016).

Serological diagnosis is generally based on the production of specific antibodies in the sera or other body fluids. IgM is the first class of antibody produced and peaks at 7–10 days following the disease occurrence, and then begins to decrease. The IgG antibody is then produced and persists for a long time after infection, facilitating the detection of infection. Therefore, an increase in IgG antibody titer over a period of about 4 weeks can usually be a sign of infection. For this reason, taking two samples from the patient at intervals of 1–4 weeks will usually be valuable in diagnosing the disease through serological tests (Vainionpää et al., 2015). Although these tests are very useful, they also have disadvantages or limitations. In certain infections, the antibody response is insufficient, or the antigens used in the test have a limited specificity, making the findings difficult to interpret. The serological response in immunocompromised patients is often too weak. In neonatal infections, the presence of maternal antibodies may make it difficult to demonstrate a response in the infant. Therefore, other diagnostic procedures should be performed in these contexts.

There are various types of serological tests that can detect antibodies or antigens in a sample. These tests can be categorized based on the following properties:

- 1) Antigen nature: including agglutination tests, precipitation tests, flocculation tests.
- 2) The function of antibodies: including neutralization test, complement fixation, hemolysis-based tests.
- 3) Labeling techniques: including ELISA, IFA, RIA, CLIA.
- 4) Blotting test: including western blotting.

In the following part we will review these tests with more details.

Agglutination tests

Agglutination occurs when an insoluble or particle antigen interacts with an antibody. A positive reaction can be detected macroscopically in a short time. However, the antigen-antibody complex may be seen with the naked eye if the complex size is large. Both IgG or IgM could be involved in the agglutination reaction. So, this test is not appropriate for distinguishing between these two classes of antibodies (Smits et al., 2000). Agglutination reactions can be divided into three categories based on the nature of antigenic determinants in antigen molecules (Oss, 2000).

- 1) Direct (active) agglutination: in this reaction, the antigenic determinants are the intrinsic component of an insoluble (or particle) antigen (Fig. 1A). For example, in bacterial agglutination assays such as the Wright and Widal tests, killed *Brucella* and *Salmonella* bacteria used to diagnose Brucellosis and Typhoid (enteric fever), respectively. In fact, in this test, dead or inactivated bacteria suspensions are used as antigens. In this test, a dilution series of the patient's serum is incubated with a defined antigen suspension. This could be done in tubes, microscope slides, or microtiter plates. A positive reaction occurs when antibodies are present in the serum. Other examples include the hemagglutination assay for identifying blood groups and the Paul Bunnell reaction for diagnosing infectious mononucleosis (Kim et al., 2015; Park et al., 2012; Tilton et al., 1988).
- 2) Indirect (passive) agglutination: in this type of agglutination, the antigen is soluble (not a particle) that binds to insoluble particles such as polystyrene latex, gelatin particles, or erythrocytes (Fig. 1B). Examples include the RA-latex test for detecting rheumatoid factor in the serum. Antibody (IgG) against rheumatoid factor is bound to a latex particle to detect rheumatoid factor in the patient's serum. Rheumatoid factor is mostly a pentameric IgM, which is found in the serum of patients with rheumatoid arthritis. The principle of the CRP-latex test is also based on indirect agglutination in which anti-CRP antibodies are attached to the latex particles. Therefore, the presence of C-reactive protein (CRP) in the patient's serum leads to a positive test (Chandra et al., 2014; Plotz and Singer, 1956).

Another example is the pregnancy test through the detection of human chorionic gonadotropin (hCG) in the urine (Fig. 1C). This test is based on agglutination inhibition, in which two forms of antigen, soluble and insoluble, compete for a specific antibody. hCG, a hormone produced during pregnancy, is a "soluble" protein and can be detected in serum or urine for pregnancy diagnosis. In this test, suspected urine is firstly mixed with specific anti-hCG antibodies. Subsequently, the latex particles coated with hCG (hCG-latex) are added to this mixture. If hCG is absent in the urine, the specific antibodies against the hormone remain free, and later, when hCG-latex particles are added, these antibodies form a complex with hCG-latex and therefore agglutination can occur. Therefore, if agglutination is observed, the test is negative. On the other hand, if hCG is present in the urine, it first binds to specific antibodies but their complex is not visible and cannot be detected because of the solubility of hCG hormone. The hCG-latex particles that are added to this complex in the next step remain free due to the lack of specific antibodies against them and therefore agglutination does not occur (Santomauro and Sciarra, 1967).

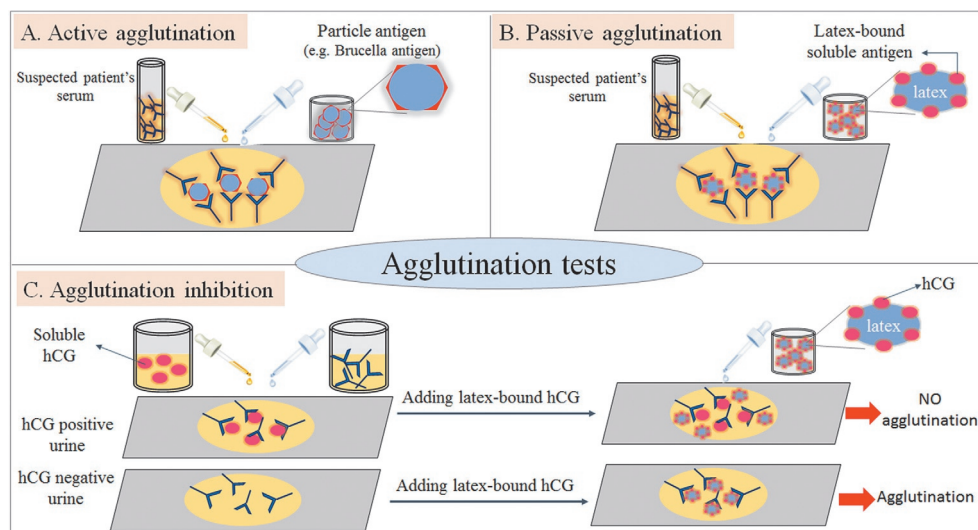


Fig. 1 Three types of agglutination tests. (A) Direct agglutination: interaction between particle (insoluble) antigen and antibodies results in the formation of large agglutinated particles which can be observed with the naked eye or by a microscope; (B) In the indirect agglutination method, soluble antigens are bound to an insoluble particle such as latex to make the reaction visible; (C) In the agglutination inhibition test, such as detection of hCG in the urine (in a pregnancy test), soluble and insoluble antigens are competing for binding to antibodies. In a pregnant person, soluble hCG in the urine binds to the anti-hCG antibodies in the first step. But, this reaction cannot be seen because of the solubility of the formed immune complexes. hCG coated latex is added in the next step. Since antibodies are bound to urine's hCG in the first step, therefore, no free antibodies remain to react with hCG-coated latex particles. As a result, the outcome of agglutination is negative. In contrast, the lack of hCG in the urine of a non-pregnant person causes the antibodies to remain free in the first step. Therefore, by adding latex particles, the agglutination occurs. hCG, human chorionic gonadotropin.

Precipitation tests

Precipitation refers to serological reactions in which the antigen is soluble. These tests can be performed on a slide, in a tube, or on agar or gel for the detection of antigen or antibody. The immunodiffusion in agar gel is the most frequent test based on precipitation. Two common types of immunoprecipitation assay include single radial immunodiffusion (SRID or Mancini method), and double immunodiffusion (DID or Ouchterlony method). In DID, antigens and antibodies are placed separately in two wells punched on the agar gel. After incubation in an appropriate condition (e.g., at 25 °C for at least 24–48 h in a humidity chamber), both antigens and antibodies are diffused in the agar and may react once they interact. In SRID, one component, for example an antigen, is placed in a pre-punched well in the agar containing antibody (antibodies are dissolved in the agar before it is placed on the slide). Therefore, the precipitation lines are formed if the antigen-antibody complex is made. An amido black staining may be used to make the precipitates more visible. These tests are rapid and simple, but they are relatively insensitive and non-quantitative. However, they can be used for the detection of either antigens (e.g., viral, bacterial) or antibodies in infectious diseases (Oss, 2000).

Flocculation tests

Flocculation refers to a reaction between a colloidal antigen and an antibody resulting in the formation of flaky and coarse immune complexes. Cardioliipin antigen is an example of a colloidal antigen that is used in the VDRL test to diagnose syphilis. Cardioliipin is prepared from beef heart extraction and is used to detect syphilitic reagin in the patient's serum. In VDRL, a suspension containing cardioliipin, lecithin, and cholesterol is added to the patient's serum. If the antibody is present in the serum, it interacts with antigenic solution and the formed immune complexes are then observed simply by a microscope. VDRL can be performed on serum or CSF samples (Vogelsang and Haaland, 1951).

Neutralization tests

Neutralization tests are based on the binding of antibodies to the pathogen or to toxins, resulting in the inhibition of their biological properties. This method can be applied to detect pathogens (with defined immunosera) or their toxins or to identify antibodies (with defined pathogens). An example the anti-streptolysin O (ASO) test for detecting antibody against streptolysin O (SO) in the patient's serum produced following streptococcal infections. The ASO test can be used for diagnosing post-streptococcal sequelae such as rheumatic fever or acute glomerulonephritis. In this test, different dilutions of the serum are incubated with a defined amount of SO enzyme (commercially ready). If the anti-SO antibodies are present in the patient's serum, they bind to the SO and neutralize it. RBC is used as an indicator to determine if the enzyme is neutralized. SO has the ability to lyse the RBCs and the neutralized enzyme would not be able to lyse RBCs (Gerber et al., 1990).

Complement fixation test (CFT)

Using the complement fixation test, the antibody titer and the amount of serum complement can be measured. Complement fixation testing is a very sensitive method based on the reaction between three components: antigen, antibody and complement. This test is performed by the macro method in tubes and by the micro method microtiter plate.

In this test, animal sensitive RBCs, usually sheep's RBCs (sRBCs) bounded to an antibody against sRBC (known as hemolysin), are used as a detector. In this procedure, the own complement of the sample should be inactivated at first. Then, known antigen and complement are added to the suspected serum. If the antigen-antibody complexes are formed, the complement system binds to them and consumed. Sensitive RBCs are then added to the mixture in the next step. If the serum is positive for antibodies, the complement is consumed in the first step because of the formation of the antigen-antibody complexes. Therefore, hemolysis-mediated complement will not occur after adding erythrocytes. As a result, if hemolysis happens, the test is likely to be negative. The titer is described as the reciprocal of the sample's highest dilution that resulted in a 50% lysis in the indicator system. In general, IgM, IgG1, and IgG3 are the most important antibodies that could be detected by CFT (Vainionpää et al., 2015).

The CFT test is a very sensitive method by which at least 0.01 µg of antibody per mL of serum can be detected. The sensitivity of this method is more than agglutination and perspiration, but less than radioimmunoassay, immunofluorescence and ELISA. This test is a common method used to serologically diagnose mycoplasma pneumoniae, fungal infections such as coccidioides, histoplasma and most viruses. However, antibodies against a variety of pathogens, including the measles virus, the mumps virus, listeria, enteroviruses, Yersinia, and the rubella virus, are no longer detectable using CFT (Amanfu et al., 2000; Stern, 1965; Shepherd et al., 1986).

This test is very complicated and tedious. Important factors are involved in the CFT, including RBC count and sensitivity, the nature and amount of hemolysin to sensitize RBCs, the hemolysis activity of complement, the ionic strength and pH of the reaction environment, and incubation time.

Hemolysis-in-gel test

In this test, antigen is attached to the surface of erythrocytes that are incorporated into the gel. The patient's serum is then added to the wells punched in the gel. Subsequently, the complement is added. If the antigen-antibody complex is formed in the well, the erythrocytes will undergo lysis. This would provide a hemolytic area around the serum sample well. This test is used to detect measles antibodies in a patient's serum. The zonal diameter increases with increasing antibody concentration. The erythrocytes remain intact if no specific antibodies are present in the serum. Both IgG and IgM antibodies can be detected in a hemolysis-in-gel test. This test could be applied for screening of infectious diseases or immunity following vaccination (Wesslén, 1978).

Enzyme-linked immunosorbent assay (ELISA)

ELISA refers to immunoassay tests using enzymes. The test is based on a colorimetric assay and is usually performed on a microtiter plate. In this procedure, enzyme-conjugated antibodies are used and the enzyme can generate measurable photometric signal following adding chromogenic substrate. Different types of enzymes are commonly used in ELISA, including alkaline phosphatase (ALP), horseradish peroxidase (HRP), and β -galactosidase. ELISA has the same sensitivity as radio immune assay (RIA) but is safer and less expensive (Aydin, 2015).

A variety of ELISA tests have been established providing the quantitative or qualitative detection of antigens or antibodies, including direct, indirect, sandwich, and competitive ELISA methods (Fig. 2).

Direct ELISA is a simple method that was first developed in 1971. This method is well suited for determining the amount of antigens or antibodies. In this regard, antigens or antibodies are coated on the surface of a microtiter plate. The photometric signal can be detected following adding enzyme-conjugated antibodies and then appropriate substrate. The generated signal is directly proportional to the amount of antigen or antibody (Fig. 2A) (Engvall, 2010).

Indirect ELISA is most commonly applied in serological testing for infectious diseases or immunological disorders. In this method, specific antigens are coated in the microtiter plates for detecting specific antibodies in the patient's serum which is added in the next step (Fig. 2B). Antigen-specific antibodies (named as primary antibodies) in the serum can bind to the coated antigens if the antibodies are present in the sample. Following washing the wells, enzyme-conjugated antibodies (specific to the primary antibodies) are added. Again, after the washing step, the appropriate chromogenic substrate is added. The photometric signal resulting from this enzymatic reaction can be measured with a spectrophotometer. Indirect ELISA is frequently used to detect pathogen-specific antibodies or antibodies against certain antigens in immune disorders. The example includes identifying the

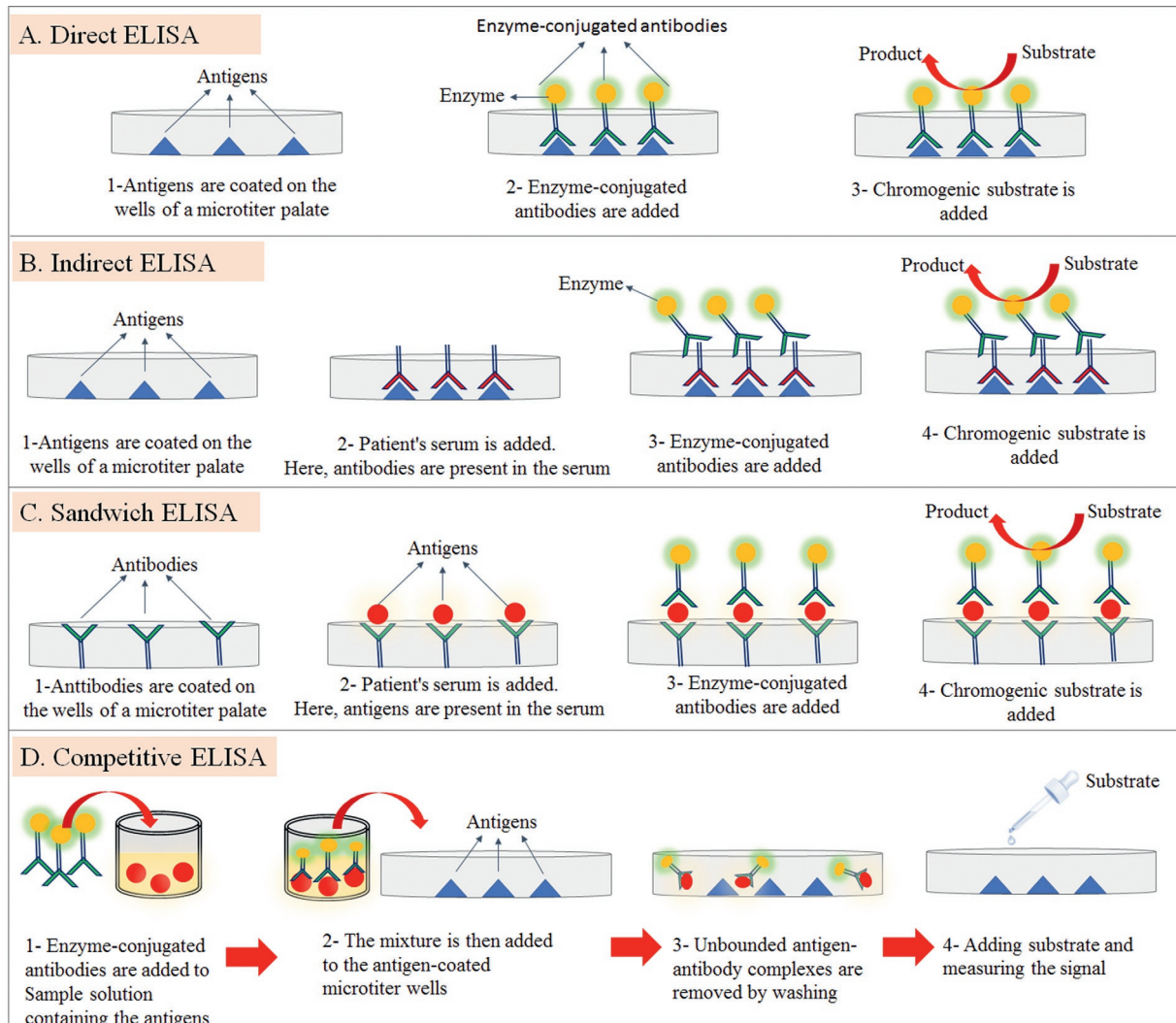


Fig. 2 Schematic diagram depicting four types of ELISA. (A) Direct ELISA; (B) Indirect ELISA; (C) Sandwich ELISA; (D) Competitive ELISA.

presence of HIV-specific antibodies in the suspected serum. For this reason, suspected serum is added to the wells of a microtiter plate coated with recombinant structural proteins of HIV (Lindström and Wager, 1978; Aydin, 2015).

Another type of ELISA technique, known as sandwich ELISA, is designed for the detection of antigens in the sample. In this method, antigen-specific antibodies (primary antibodies) are coated on a microtiter plate (Fig. 2C). By adding the unknown sample in the next step, the antigen in the sample binds to primary antibodies (if the sample is positive). Enzyme-conjugated antibodies are then added in the next step. These antibodies are specific to the primary antibodies. By adding a substrate, the photometric signal can be measured with a spectrophotometer (Kato et al., 1977).

Another procedure to assess the level of antigens or antibodies in the sample is through competitive ELISA, in which the antibodies or antigens in the patient's sample compete with a defined, enzyme-conjugated antibody or antigen over a binding site. In this method, the enzyme-conjugated antibodies are first added to the sample solution containing the antigen (Fig. 2D). This antibody-antigen mixture is then added to the antigen-coated microtiter wells. The high presence of antigen in the sample causes less free antibody to remain to bind to the coated antigens of the well. As a result, the analyte concentration has an inverse relation to the measurement signal (Yorde et al., 1976).

Another technique is designed to increase the sensitivity of the ELISA method. Enzyme-linked fluorescent assay (ELFA) is developed in which a fluorescence signal (e.g., 4-Methylumbelliferyl phosphate) is detected instead of a photometric color signal. The ELFA procedure time is quicker than ELISA and it can detect up to 10–18 mol/assay compared to ELISA that can detect only up to 10–16 mol/assay (Yolken and Stopa, 1979).

Immunofluorescence assay (IFA)

The Immunofluorescence assay can detect either antigen or pathogen-specific antibodies (Bayer et al., 2001). In this procedure, antibodies are labeled with a fluorochrome dye such as fluorescein isothiocyanate (FITC), rhodamine, and phycoerythrin. Therefore, immune complexes containing these fluorescence-labeled antibodies can be detected by colored light emitted (after being excited by light of the appropriate wavelength). The emitted light can be seen with a fluorescent microscope equipped with ultraviolet light and suitable fluorochrome filter systems. Each fluorochrome absorbs a wavelength of light and emits light at a higher wavelength. Fluorescein absorbs blue light (490 nm) and emits a fluorescent yellow-green (517 nm). Rhodamine absorbs yellow-green light (515 nm) and emits a red fluorescent dye (546 nm). Rhodamine can be used in two-color immunofluorescence tests because it emits fluorescence at a higher wavelength than fluorescein. For example, by binding fluorescein to one antibody and rhodamine to another antibody, two membrane antigens can be observed on one cell simultaneously. Phycoerythrin absorbs slightly blue-green/yellowish light (565 nm) and emits slightly orange-yellow light (573 nm) (Mullins, 2010).

Using fluorescent antibodies, the antigen or antibody of interest can be identified by a direct or indirect immunofluorescence method, respectively. Direct immunofluorescence (DIF) is a one-step histological staining procedure in which tissue antigens (fixed in a solid phase, mostly slides) can be recognized directly by adding fluorochrome-labeled antibodies. The signal is observed if the antigen-antibody complex is formed (Vainionpää et al., 2015; Aoki et al., 2010). Indirect immunofluorescence (IIF) is a two-step serological technique in which a labeled antibody is used to detect a specific antibody against an antigen in the serum. For this reason, suspected serum is added to a glass slide with antigen-coated fields, followed by adding specific labeled-antibodies. If the serum is positive for antibodies, the formed complex can be detected by a fluorescent microscope (Aoki et al., 2010; Huang et al., 2012). Pathogens such as *Toxoplasma gondii* can be detected by this method (Bernal-Guadarrama et al., 2014). This procedure is also commonly used for detecting pathogen-specific antibodies in autoimmune diseases such as systematic lupus erythematosus (SLE) (Ghosh et al., 2007). The example includes identifying the antinuclear antibodies (ANA) in the serum. ANA may be produced against different nuclear antigens in the serum of patients with a variety of rheumatic and nonrheumatic disorders. Another example is the fluorescent treponemal antibody absorption (FTA-Abs) test (Hunter, 1975). Treponema-specific antigens are coated on the slide, then the unknown serum is added. After washing and adding the labeled antibody, the result can be seen by a microscope. This test is usually performed if the initial screening tests (e.g., rapid plasma regain (RPR), venereal disease research laboratory (VDRL)) for syphilis are positive, so that this test can help confirm whether the results of screening tests are accurate. In general, the IIF has more sensitivity compared with DIF. In another technique based on IF staining, known as double IF, the presence of two different antigens on a cell can be identified.

Radio immunoassay (RIA)

RIA is an immunoassay using radiolabeled antibodies or antigens. Iodine isotopes such as ^{125}I , ^{131}I or tritium can be used for labeling. This method has high sensitivity but needs specialized equipment (e.g., gamma counter) for detecting radioactive signals. Furthermore, it requires special handling and disposal of radioactive waste. In this method, the unlabeled and labeled antigens compete for interaction with antibody molecules to form an antigen-antibody complex. The formed complex is then separated and precipitated with anti-immunoglobulin. The radioactivity of the precipitate indicates the bound antigens that is inversely correlated to the unlabeled antigen concentration (Goldsmith, 1975).

Chemiluminescence immunoassay (CLIA)

The basis of the CLIA method is similar to that of ELISA, except that CLIA substrates can generate light emission in the presence of an enzyme, providing a more sensitive process compared to ELISA. Substrates such as isoluminol or acridinium ester produce the luminescence signal in the presence of hydrogen peroxide and enzyme. The electro-chemiluminescence immunoassay (ECLIA),

another type of CLIA, uses electrical current for oxidizing substrate. CLIA and ECLIA methods have higher sensitivity compared to ELISA and ELFA and have a shorter analysis time (Liu et al., 2018).

Immunoblotting method

Immunoblotting or western blotting (WB) is a technique recognizing specific antibodies or antigens (e.g., pathogen antigens) in the sample. The presence of a certain protein and its molecular weight can be identified by WB. The method is based on a combination of gel electrophoresis and formation of the antigen-antibody complex (Ni et al., 2017; Magi and Liberatori, 2005). The procedure involves four important steps, as follows:

- 1) Protein separation by SDS-PAGE: the protein mixture is separated by SDS-PAGE based on the molecular weight of proteins.
- 2) Protein transfer from the gel to a nitrocellulose membrane by electrophoresis.
- 3) Incubation of the nitrocellulose membrane with serum sample: antigen-specific antibodies in the serum (primary antibodies) bind to the antigens on the nitrocellulose membrane.
- 4) Adding labeled secondary antibodies (specific to the primary antibodies) and detecting signals using either a chromogen or an X-ray film.

The antigen-antibody complexes can generate a visible band on the carrier membrane as a result of a dye reaction. By using known antigens, blot methods reveal a high specificity for detecting antibodies (e.g., HIV antibodies). However, WB has lower sensitivity compared to ELISA, ELFA and CLIA methods depending on the pathogen specificity. Therefore, it can be used as a confirmatory test (e.g., in borreliosis diagnosis) or as an additional test (e.g., toxoplasmosis, CMV, EBV diagnosis).

Nucleic acid-based identification methods

The principle of nucleic acid-based methods is the detection of specific DNA or RNA sequences of the microorganism. Nucleic acid-based approaches provide precise and sensitive identification and diagnosis of pathogens causing infectious diseases, as well as knowledge of the disease's epidemiology. These tests are rapid and have high specificity and sensitivity and can be used for the detection of all types of microbes. The recent development of the multiplex assay provides the detection of several different antigens in one analytical procedure. However, these tests are less sensitive or less specific than the single-target tests. In the following section, the most important nucleic acid-based methods are reviewed (Mothershed and Whitney, 2006; Mianzhi and Shah, 2017; Yu et al., 2012).

Polymerase chain reaction (PCR)-based amplification

PCR is an in vitro enzymatic reaction that enables large amounts of a particular DNA fragment to be generated. When compared to serologic tests, this method is more accurate and faster (Garibyan and Avashia, 2013; Ashford et al., 1995). Serologic tests are limited because of the window period required for production of sufficient levels of antibodies, so false-negative results can occur during this time. PCR can be applied even in the early phase of infection to detect the presence of a pathogen. For example, tuberculosis bacteria multiply slowly, making diagnosis via culture or serological methods time-consuming. In this condition, the PCR technique allows rapid amplification of the pathogen genes, leading to increased detection sensitivity and decreased detection time. Another example that highlights the importance of PCR in clinical diagnosis is Lyme disease, caused by a tick-borne spirochete. Using culture for detecting these bacteria is approximately impossible since they are too small to be seen by a microscope. Other examples are detecting Influenza virus, blood screening for human immunodeficiency virus (HIV), hepatitis B, hepatitis C and syphilis. Three important steps of PCR are as following (Fig. 3):

- 1) Denaturation: the use of high temperature at this step causes the double-stranded DNA to become single-stranded.
- 2) Annealing: as the temperature is reduced, the primers recognize and bind to the target DNA.
- 3) Extension: at this step, the complementary strand is made by a DNA polymerase enzyme at the appropriate temperature.

These three steps are generally repeated for 30–40 cycles. Importantly, the specificity of the designed primer to the target sequence determines the specificity of the PCR. After the reaction is over, the amplified DNA can be visualized using gel electrophoresis. Although computer software can estimate the amount of amplified DNA from the intensity of the band on the gel, PCR is not suitable for quantitative studies. For this purpose, quantitative PCR (qPCR) or real-time PCR (RT-PCR) has been developed. The PCR reaction can be monitored in RT-PCR through the emission of fluorescence during the reaction. These signals are recorded by the RT-PCR machine and are therefore traceable at the same time as the reaction is done. When compared to PCR, RT-PCR has a much higher degree of accuracy (Freeman et al., 1999).

Isothermal amplification

Isothermal nucleic acid amplification is a simple process that uses a constant temperature for the amplification of a nucleic acid target sequence instead of thermal cyclers required for PCR (Wong et al., 2018). Several isothermal amplification methods have been developed based on the mechanism of the reaction. Nucleic acid sequence-based amplification (NASBA) and Loop-mediated isothermal amplification (LAMP) are the most widely used isothermal methods. NASBA is an isothermal and continuous amplification based on RNA transcription, providing accurate results with high sensitivity (Compton, 1991; Morré et al., 1996).

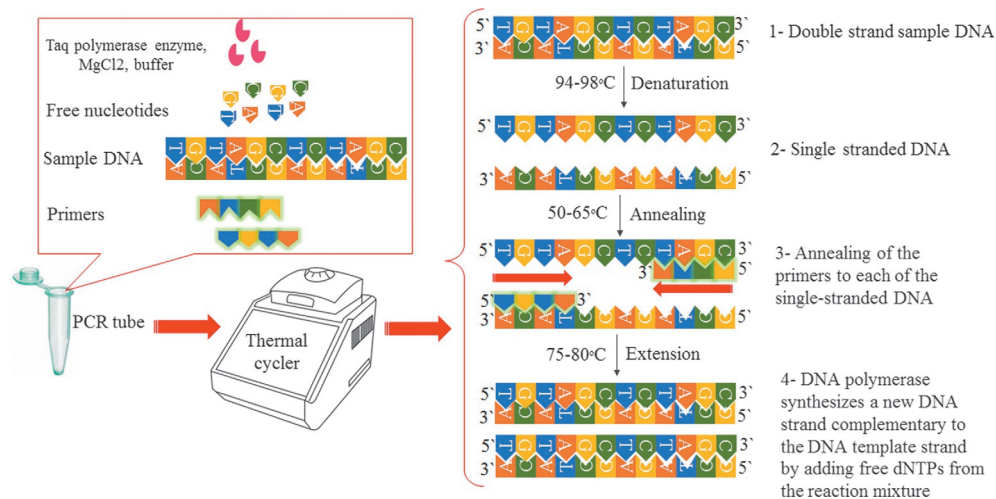


Fig. 3 Schematic diagram of the polymerase chain reaction (PCR). These three steps are generally repeated for 30–40 cycles. Here, only 1 cycle is depicted.

It can be easily done in a water bath without the need for a thermocycler. The reaction takes place in the presence of a mixture of reverse transcriptase, RNA polymerase, ribonuclease-H, and two specially designed DNA oligonucleotide primers. A 5' extension of the forward primer includes the promoter sequence for bacteriophage T7 DNA-dependent RNA polymerase. A 5' extension of the reverse primer includes a sequence for binding a probe tagged with ruthenium-based electrochemiluminescent (ECL) that can be detected by a chemiluminescence detector (Lau et al., 2006). Under ideal conditions, the yield of amplification is nearly 10^{12} -fold, compared to 10^9 -fold amplification by PCR (Boom et al., 1990).

NASBA has been used extensively in the identification of a variety of viruses, including HIV-1, avian Influenza virus, Newcastle disease virus, Simian immunodeficiency virus, and Respiratory syndrome virus. This method can also be used in the detection of cytokines, bacteria (e.g., *Salmonella enterica*), fungi, and parasites (Romano et al., 1996, 2000; Collins et al., 2003; Lau et al., 2006).

LAMP is another isothermal amplification method conducted in the presence of a DNA polymerase (with a function for strand displacement) as well as two sets of primers (four different primers) (Zhao et al., 2015). Since amplification only occurs when all six sequences of a target DNA are accurately identified by the primers, the LAMP assay has a high specificity. However, the difficulties in developing usable primers restrict the practical application of LAMP. The amplification process continues in a cyclical manner. The end product is stem-loop DNA with multiple inverted copies of the target sequence. Positive LAMP reaction products can be identified using a variety of detection methods, such as incorporation of SYBR Green I stain or using polyethylenimine (PEI). This technique has been used in the detection of several viruses (e.g., Noroviruses, SARS-CoV, H5N1, avian Influenza subtypes, Influenza subtype 1, Influenza subtype 3, and Influenza B virus) and bacteria (e.g., *Salmonella*, *Escherichia coli* and *Mycobacterium tuberculosis*) (Hara-Kudo et al., 2005; Hill et al., 2008; Iturriza-Gómara et al., 2008; Poon et al., 2004; Imai et al., 2006; Geojith et al., 2011).

Microarray/chip-based technologies

The development of microarray/chip-based technology enables the analysis of samples with a large number of sequences in a single test (Raghavachari, 2013; Marzancola et al., 2016). Pathogen screening involves immobilizing the expressed sequences of various pathogens on a filter to determine which pathogens are present in the suspected samples. Therefore, a single procedure allows the detection of multiple pathogens. The basic principle of microarrays is the hybridization between two strands of DNA. In this procedure, a high number of specific DNA sequences known as probes or oligos are immobilized on a solid surface that are used to hybridize a nucleic acid of interest in the sample. The sample's nucleic acid is purified and labeled, typically with a fluorescent tag. The labeled samples are then added to the pinholes of the microarray. After hybridization has occurred, all nonspecific binding is washed off. Finally, the microarray is scanned by a machine through a laser to excite the dye and measures the emission levels with a detector.

Microarray-based methods have been used widely in screening clinical samples for viruses. This technique is also applied in single nucleotide polymorphisms (SNPs) analysis and in cancer diagnosis, such as detecting P53 mutations, and distal metastasis of lymph-node-negative primary breast cancer (Ahrendt et al., 1999; Wang et al., 2002, 2005; Chiu et al., 2006; Schick et al., 2011).

Non-nucleic acid-based identification methods

It is essential to identify an organism after it has been isolated by culture. The importance of identification is in the management of infectious diseases, such as choosing a better drug for treatment. Rather than genetic recognition, non-nucleic acid-based identification approaches depend on phenotypic characteristics of organisms, including functional or morphologic. There are several biochemical tests available, each of which is limited to species of a certain type (e.g., aerobic or anaerobic bacteria). However, two techniques, including chromatographic methods and mass spectrometry, are widely used for identifying pathogens.

High-performance liquid chromatography (HPLC) or gas chromatography can be used to isolate and identify microbial components or products. Typically, recognition is accomplished by comparing an organism's fatty acids to a database. In general, aerobic and anaerobic microbes, mycobacteria, and fungi can all be identified using chromatographic techniques. The accuracy of the test is determined by the circumstances under which the specimen was cultured and the accuracy of the database, which could be unreliable or inadequate (Larsson and Mårdh, 1977; Lima et al., 2018; Emonet et al., 2010).

Pathogens have distinct proteins, and the relative concentration and mass of each protein may often be used to classify a microorganism. Different proteins with different masses in a sample can be detected using mass spectrometry. A type of mass spectrometry known as matrix-assisted laser desorption/ionization-time of flight (MALDI-TOF) is currently being applied to recognize mycobacteria, yeasts, molds, and viruses (Patel, 2015).

Diagnostic laboratory tests: Focusing on immunological disorders

Autoimmune disorders

Autoimmunity is defined by a lack of immune homeostasis, which causes the organism to develop an abnormal reaction to its own tissue. Laboratory tests can be helpful in disease management and determining disease severity. The development of self-reactive T cells, autoantibodies, and inflammation are common features of autoimmune diseases. Diagnosing autoimmune disorders is complex and several laboratory tests are usually needed. These laboratory diagnosis tests include specific and non-specific tests such as complete blood count (CBC), evaluation of acute phase reactants and inflammatory markers, serological tests, cytokine analysis, and immunologic studies (Castro and Gourley, 2010). Laboratory diagnostic tests for evaluating autoimmune disorders are summarized in Table 1.

Inflammation-related markers

Inflammatory markers are serum proteins that are produced by the liver in response to inflammation. Pro-inflammatory cytokines such as IL-1, IL-6, and TNF-alpha produced by immune cells in the inflammatory condition play a significant role in stimulating the production of acute phase proteins such as C-reactive protein (CRP), fibrinogen, and haptoglobin. Inflammatory markers cannot be used to diagnose inflammation, although they can reveal abnormalities in a variety of conditions such as infections, autoimmune diseases, and malignancies (Gruys et al., 2005).

Erythrocyte sedimentation rate (ESR) is a test applied for the evaluation of the RBC's quantity. ESR is performed by analyzing the RBC sedimentation rate in a tube at a defined time. While ESR is a non-specific test and cannot be considered as a diagnostic test, it can be useful in controlling disease severity and therapeutic response, and can reflect the presence of inflammatory or infectious stress (Lahita and Weinstein, 2007).

CRP is an acute phase protein which can be induced under the influence of pro-inflammatory cytokines. This protein has an important activity in inducing phagocytosis through pathogen opsonization as well as complement activation. Since serum CRP concentrations shift faster than ESR, CRP could be a better indicator of current inflammation. On the other hand, in contrast to ESR, CRP is a stable serum protein that is not influenced by other serum components. CRP can be measured quantitatively and qualitatively (e.g., by CRP-latex described in the serological methods). The level of inflammation is proportional to the level of CRP. Levels of less than 0.2 mg/dL are regarded as normal, whereas levels greater than 1.0 mg/dL are indicative of inflammation and/or infection (Gruys et al., 2005; Lahita and Weinstein, 2007).

Ferritin is another serum protein that its increased levels may suggest acute or chronic sepsis, inflammation, or cancer. Ferritin is frequently increased in diseases such as systemic-onset juvenile idiopathic arthritis and lymphohistiocytosis. Other inflammatory markers include Ceruloplasmin, Fibrinogen, and Haptoglobin that can be increased during inflammation and autoimmunity disorders (Breda et al., 2010; Koulaouzidis et al., 2009).

Autoantibodies

The existence of autoantibodies in a patient does not imply a diagnosis of an autoimmune disorder. The signs and symptoms of the patient should also be carefully considered. Currently, ELISA (described in serological methods) is frequently used to detect autoantibodies in a suspected patient (Eggert et al., 2010).

"Rheumatoid factor" (RF) is an autoantibody produced against the FC region of IgG. RF is used to evaluate patients with rheumatoid arthritis (RA) with a sensitivity and specificity of about 70%. The RF can be any class of antibodies. However, most methods identify the IgM rheumatoid factor. It should be considered that approximately 15% of rheumatoid arthritis patients are negative for RF. On the other hand, RF may be positive in some healthy individuals. Although RF is considered as a non-specific test, patients with positive RF may experience progressive arthritis, extraarticular manifestations, and secondary Sjögren's syndrome. RF can be detected in other autoimmune diseases such as SLE, Sjögren's syndrome, and cryoglobulinemia. Identifying another autoantibody known as anti-citrullinated protein (anti-CCP) is also valuable in diagnosing RA patients (with about 95% specificity). Anti-CCP can be positive in psoriatic arthritis, Sjögren's syndrome, juvenile idiopathic arthritis (JIA), and inflammatory myopathies (Lahita and Weinstein, 2007; Klippel et al., 2008).

"Anti-nuclear antibody" (ANA) is another autoantibody which is mostly used in the diagnosis of SLE. However, ANA can be found in other diseases, including RA, scleroderma, Sjögren's syndrome, mixed connective tissue disease, and polymyositis/dermatomyositis. The immunofluorescence method is frequently used for detecting ANA in patients' serum. ELISA technique is

Table 1 Laboratory diagnostic tests for evaluating autoimmune diseases.

<i>Autoimmune disorder</i>	<i>Laboratory diagnostic tests</i>
Rheumatoid arthritis (RA)	• CRP and ESR: non-specific, initial screening, indication of inflammation • RF: not specific, present in about 80% of patients with RA, associated with poor prognosis • Anti-CCP: more sensitive and specific compared to RF, present in about 80% of patients with RA • ANA: about 60% specificity and 45% sensitivity in patients with RA • HLA-Typing: HLA-DRB1 variations
Systemic lupus erythematosus (SLE)	• CBC: anemia, low WBC or platelet count maybe seen • ANA: positive in about 100% of the patients with SLE, higher sensitivity (about >95%) • Anti-dsDNA, higher specificity (about 95%), usually associate with active disease • Anti-Extractable Nuclear Antigens: Anti-sm (high specificity), anti-RNP, Anti-SSA (anti-Ro), Anti-SSB (anti-La) • Urinalysis in renal involvement: hematuria, proteinuria, present WBC • Complement assay: decreased level of C3 and C4, decreased CH50 • Kidney biopsy
Goodpasture's syndrome	• Urinalysis: indicates proteinuria, dysmorphic RBCs, WBCs, and RBC cellular and granular casts • Anti-GBM • Kidney biopsy
Sarcoidosis	• The levels of serum ACE: elevated levels may be seen, but this test is non-specific • The levels of Vitamin D: increased levels of 1, 25(OH)2D3 vitamin D are commonly seen • The levels of SAA: increased levels of SAA are commonly seen
Churg-Strauss syndrome	• CBC: indicates a peripheral eosinophilia • ESR and CRP: elevated levels are seen • ANA and a p-ANCA against MPO • A biopsy of the involved organ: for evaluation of vasculitis and granuloma formation
Wegener's granulomatosis (WG)	• ESR and CRP: elevated levels are seen • C-ANCA-PR3 • HLA-Typing: WG can be associated with HLA-DR4, DR13 • Urinalysis: proteinuria, hematuria, cellular casts • The levels of serum albumin: hypoalbuminemia
Myositis	• CBC: May show leukocytosis or thrombocytosis • Muscle biopsy • MSA: Anti-tRNA synthetase antibodies (anti-JO-1 antibodies), Anti-SRP, Anti-PM/ScI, Anti-Mi-2 • HLA-Typing: myositis can be associated with HLA-DRB1 *0301 and HLA-DQA1 *0501 • The levels of CK, AST, ALT and LDH
Autoimmune hemolytic anemia (AIHA)	• Peripheral blood smear: polychromasia, spherocytosis, fragmented and nucleated erythrocytes Reticulocytosis, erythrophagocytosis by monocytes in severe cases • The levels of haptoglobin: decreased levels are seen • The levels of LDH: increased levels are seen • DAT: evaluation of immunoglobulin and/or complement proteins bound to the patient's erythrocytes
Antiphospholipid syndrome	• CBC: thrombocytopenia • aPL
Idiopathic Thrombocytopenia	• Thrombocytopenia • human platelet antigen—1a antibodies

CRP, C-reactive protein; ESR, Erythrocyte sedimentation rate; RF, Rheumatoid factor; CCP, Cyclic citrullinated peptide; ANA, Anti-nuclear antibody; HLA, Human leukocyte antigen; CBC, Cell blood count; WBC, White blood cell; SLE, Systemic lupus erythematosus; dsDNA, Double stranded DNA; Anti-sm, Anti-smith; anti-RN, Anti-ribonucleoprotein; Anti-SSA, Anti-Sjögren's-syndrome type A; Anti-SSB, Anti-Sjögren's syndrome type B; CH50, Hemolytic complement; Anti-GBM, Anti-glomerular basement membrane antibodies; ACE, Angiotensin converting enzyme; SAA, Serum amyloid A; ANCA, Antineutrophil cytoplasmic antibodies; p-ANCA, Perinuclear-ANCA; c-ANCA, Cytoplasmic-ANCA; MPO, Myeloperoxidase; PR3, Proteinase 3; WG, Wegener's granulomatosis; MSA, Myositis-specific antibodies; Anti-PM/ScI, Antibody to nucleolar granular component; Anti-SRP, Antibody to signal recognition protein; CK, Creatinine kinase; AST, Aspartate aminotransferase; ALT, Alanine transaminase; LDH, Lactate dehydrogenase; DAT, Direct anti-globulin test; aPL, Antiphospholipid antibody.

also used with high sensitivity but low specificity. ANA is also found in nonrheumatic disorders such as autoimmune hepatitis, Hashimoto's thyroiditis, primary pulmonary hypertension, Graves' disease, and primary autoimmune cholangitis (Sur et al., 2018; Grygiel-Górniak et al., 2018). Furthermore, ANAs are also associated with several infections, such as *Mycobacterium tuberculosis*, *Treponema pallidum*, *Orientia tsutsugamushi*, *Escherichia coli*, *Bartonella henselae*, and human immunodeficiency virus (Im et al., 2020).

Other antibodies that are important and highly specific in diagnosing SLE are "anti-double stranded DNA" (anti-dsDNA). Although anti-dsDNA was previously detected by radioimmunoassay, IFA and ELISA are more commonly used currently. Crithidia luciliae, a flagellated protozoa with a dsDNA-containing small organelle called a kinetoplast, is used as a target antigen by the IFA. Semiquantitative detection of antibodies against dsDNA is accomplished by detecting IgG attached to the kinetoplast. In ELISA, the dsDNA is coated on a microtiter plate and the bound IgG is detected after adding the suspected serum (Lahita and Weinstein, 2007; Mummert et al., 2018; Conrad et al., 2009).

“Antineutrophil cytoplasmic antibodies” (ANCA) are produced against the cytoplasmic granules of neutrophils. The immunofluorescent patterns can be useful in diagnosing various ANCA associated vasculitis syndromes. Cytoplasmic pattern (cANCA) is frequently found in Wegener’s Granulomatosis (WG), microscopic polyangiitis (MPA) and Churg-Strauss syndrome. Perinuclear pattern (pANCA) can be seen in MPA as well as in RA, inflammatory bowel disease (IBD), JIA, and SLE. The vasculitis type and disease severity can also be identified by detection of proteinase 3 (PR3) and myeloperoxidase (MPO). For example, MPO-cANCA is particularly associated with WG. MPO-pANCA is positive in MPA and less often for Churg-Strauss. It should be considered that ANCA can also be positive in some conditions, such as during thyroid medication (e.g., PTU) and other immune disorders. Tissue biopsy can be helpful for confirming patients suspected of vasculitis (Schönermarck et al., 2001; Csernok and Gross, 1995).

“Anti-extractable nuclear antigen” (anti-ENA) are a group of antibodies against extractable antigens, including SSA (Ro) and SSB (La) nuclear antigens, Smith (Sm) antigen, and ribonuclear protein (RNP) or U1RNP. Anti-SSA and anti-SSB antibodies are usually seen together in Sjögren’s syndrome patients and some patients with SLE.

Anti-Sm is particularly specific for SLE; However, it is only present in 25% of SLE patients. Antibodies to U1 ribonucleoproteins have been detected in several autoimmune diseases such as SLE, systemic sclerosis, and in patients with mixed connective tissue disease (Agarwal et al., 2009).

Autoantibodies known as anti-JO-1, anti-Mi-2, anti-SRP, and anti-PM/Scl have been detected in the autoimmune inflammatory myopathies (IIM) with high specificity. Anti-JO-1 antibodies are produced against histidyl-tRNA synthetase and are particularly seen in anti-synthetase syndrome, an inflammatory myopathy. Anti-Mi2 antibodies can target a nuclear complex involved in transcription. These autoantibodies are specific for diagnosing dermatomyositis. Anti-signal recognition particles (SRP) are antibodies against a cytoplasmic RNA protein involving 7S RNA and 6 other proteins. Anti-SRP has been detected in the serum of polymyositis patients. These antibodies are considered as serological markers for identifying necrotizing myopathy. Anti-PM/Scl is a type of antinuclear antibody associated with idiopathic inflammatory myositis-systemic sclerosis overlap syndromes (also called scleromyositis or sclerodermatomyositis). These types of autoantibodies have also been found in patients with polymyositis, dermatomyositis and systemic sclerosis without features of overlap syndromes (Satoh et al., 2017).

Autoantibodies named as antiphospholipid autoantibodies (aPL) or lupus anticoagulant (LAC) or anti-cardiolipin (aCL) are produced against phospholipids as well as phospholipid-binding proteins. These antibodies have been found in patients with anti-phospholipid antibody syndrome (APS) as well as SLE patients. aPL antibodies may also be detected in healthy individuals (McIntyre et al., 2003; Coín et al., 2015).

Complement activity

The complement system is an enzymatic cascade consisting of several proteins in the serum. Complement activation can be considered as an important pathway in defending against pathogens. Meanwhile, the levels of complement components in the serum can be used to identify disease activity. The methods for measuring complement components include nephelometry and ELISA for detecting components such as C3, C4 and factor B. The hemolytic activity of complement can be assayed using CH50 and AH50 which analyze the functional activity of the classic and alternative pathways, respectively. In diseases in which the immune complex is formed in tissues, such as glomerulonephritis in patients with SLE, the serum levels of complement’s proteins decrease. In contrast, elevated levels of C3 and C4 in the acute phase of the disease can suggest an inflammatory disorder. A decreased level of complement components has been found in nonrheumatic disorders such as post-streptococcal glomerulonephritis or bacterial endocarditis (Conigliaro et al., 2019; Boackle, 2003).

Cytokines

Cytokines are glycoproteins produced by several immune and non-immune cells, playing an important role in immune function and the development of inflammation. Flow cytometry and ELISA are laboratory techniques used to evaluate cytokine levels. For this reason, sandwich ELISA is frequently used by commercial kits. Although evaluation of the gene expression levels of cytokines can be applied by molecular tests, particularly Real-Time PCR, this process is usually performed in research laboratories.

Assessing pro-inflammatory cytokines, especially IL-1, IL-6, and TNF-alpha can be helpful in detecting an inflammatory disorder. Since the level of pro-inflammatory cytokines is increased in autoimmune disorders, targeting these molecules is used as an effective therapeutic approach. For example, targeting TNF-alpha through monoclonal antibodies (mAbs) can significantly reduce disease severity in patients with RA. Using mAbs, including anti-TNF, anti-IL-1 and anti-IL-6 therapies, is currently being used in several immune disorders (O’Shea et al., 2002; Moudgil and Choubey, 2011).

Human leukocyte antigen (HLA) typing

The association between HLA or Major Histocompatibility Complex (MHC) and immune disease susceptibility has been described in many studies. HLA-A, -B and -C are the classic types of MHC class I molecules, and HLA-DR, HLA-DQ, and HLA-DP are MHC class II molecules. It has been found that many rheumatic disorders are associated with HLA class I and II. For example, HLA-B27 is highly associated (about 90–95%) with susceptibility to ankylosing spondylitis. HLA-DR1 and HLA-DR4 are associated with an increased risk of polyarticular juvenile idiopathic arthritis (JIA). In Caucasian populations, HLA-DR3 and HLA-DR2 are associated with lupus, while HLA-DRB1 is associated with an increased risk of rheumatoid arthritis (Troshina et al., 2020).

HLA-typing is also considered an important test before organ transplantation (especially bone marrow transplantation) to identify which people can provide the safest tissue transplants. In this regard, the best potential donors have HLAs that closely match the recipient’s HLA profiles. Several methods can be used to detect HLA types, including serological- and molecular-based methods (Wiebe and Nickerson, 2020).

Serological typing (traditional method) is frequently used in many laboratories for identifying HLA-A and HLA-B with low resolution. This method is performed using a complement-dependent cytotoxicity (CDC) test or micro lymphocytotoxicity assay. In this procedure, a panel of sera containing HLA-specific alloantibodies is added to a well of a microtiter Terasaki plate consisting of the patient's lymphocytes. Rabbit serum containing complement is added after a short incubation time. The lymphocytes that have been recognized by a specific alloantibody undergo lysis, allowing the fluorochrome ethidium bromide to enter. The lysed cells can simply be identified by microscopy (Saito et al., 2014).

Molecular testing provides HLA-typing with more resolution using the PCR technique. These tests including sequence-specific primer (SSP) and sequence-specific oligonucleotide (SSO) are very useful and both of them can apply either for generic (low resolution) and allelic (high resolution) typing. SSP provides identification of HLA-A, -B, -C, -DRB1, -DRB3, -DRB4, -DRB5, and -DQB1 alleles. SSP typing includes using primers that are complementary to specific HLA allele sequences (e.g., HLA*02:03) or a set of identical alleles (e.g., all A*02 alleles). The presence of the product is then detected by agarose gel electrophoresis. This method is performed rapidly and is ideally appropriate for small numbers of samples (Madden and Chabot-Richards, 2019; Kulcsarova et al., 2000).

Sequence-specific oligonucleotide (SSO) was first described for HLA class II and then for HLA class I typing. In this method, locus-specific amplification of the genomic DNA segment is initially performed by PCR. After that, the amplified DNA is blotted onto a solid support, normally a nylon membrane, and hybridized with a series of sequence-specific oligonucleotide probes (SSOP) (direct hybridization). The probes are bound to the fluorochromes, allowing their detection by chemiluminescence. In indirect hybridization, SSO probes are attached to a solid phase and then hybridized with a labeled PCR product. The SSO method is very suitable for the typing of large sample numbers. However, a method known as reverse-SSO has been developed for testing low sample numbers. In this method, a pre-made SSOs bound membrane is used. Sample DNA is specifically amplified using specific primers for certain HLA locus. Amplified DNA is labeled (e.g., with biotin) and then added to the SSOs bound membrane. Finally, the hybridization result is detected using a colorimetric or chemiluminescent detection system. Recently, a reverse-SSO system is developed based on Luminex technology. All methods, including SSP, SSO and reverse-SSO, are all appropriate for application in research and also in routine tissue typing laboratories (Madden and Chabot-Richards, 2019; Kulcsarova et al., 2000).

Another technology known as Sequence-based typing (SBT) is a comprehensive molecular-based method with high resolution. This technique is based on the sequencing of amplified DNA. Both alleles for each HLA locus are amplified and then sequenced. Although PCR-SSO is able to detect single-base variations in DNA sequences within two alleles, it is less likely to detect new undefined alleles unless the variation occurs at the particular site detected by the probe or the restriction enzyme. Furthermore, this method is more expensive compared to other methods (Voorter et al., 2014).

Immunodeficiency disorders

Immunodeficiency disorders are referred to a group of diseases in which the immune system is compromised because of defects in innate immune system (e.g., complement pathway, phagocytosis) or adaptive immune system (e.g., cellular and humoral immunity). Primary immunodeficiencies are determined genetically and they are usually revealed during infancy or childhood. Secondary immunodeficiencies can result from other disorders or conditions such as diabetes, radiation therapy, chemotherapy, bone marrow ablation before transplantation, etc. There are a variety of diagnostic laboratory tests for diagnosing defects in the immune system (Rezaei et al., 2009; Raje and Dinakar, 2015). Laboratory diagnostic tests for evaluating primary immunodeficiencies are summarized in Table 2. We will review these tests in the following sections.

Initial screening

In general, recurrent infections are a common symptom of immunodeficiency. The patient can experience up to 10 respiratory infections/year. Therefore, a careful history of the patient to assess infectious diseases and other complications is crucial to diagnosis of an immunodeficiency disorder. In this regard, manifestations such as failure to thrive (FTT), upper and lower respiratory tract infections (e.g., pneumonia, sinusitis, bronchitis), and chronic diarrhea can be considered as frequent manifestations importantly in a patient with primary immunodeficiency. Therefore, it is essential to evaluate clinical manifestations along with laboratory diagnostic assays. Initial laboratory tests for screening such patients should include complete blood counts (CBC) and differentials (platelet volume, absolute lymphocyte count, neutrophil and eosinophil counts), immunoglobulin levels (IgG, IgM, IgA), lymphocyte subpopulations, vaccine titers, and complement assessment (e.g., CH50, AH50).

CBC results can detect abnormalities in one or more cell types, which can be a clue to detect a specific disorder. These abnormalities include anemia, neutropenia, thrombocytopenia, and lymphopenia. For example, since T cells account for about 70% of circulating lymphocytes, lymphopenia could suggest a T-cell associated deficiency. A patient with persistent leukocytosis, especially between infections, may suspect a leukocyte adhesion deficiency. Wiskott-Aldrich syndrome can be initially screened by thrombocytopenia. On the other hand, decreased levels of immunoglobulins can suggest antibody deficiency. However, these assessments are often inadequate to make a diagnosis. Therefore, several complementary tests should be performed when the screening tests and clinical manifestations are in favor of an immunodeficiency disorder. These tests are usually performed to assess immune cell function (Locke et al., 2014; Eghbali et al., 2015).

Humoral immunity deficiency

When a humoral immunity-related immunodeficiency is suspected, quantitative immunoglobulin analysis (IgG, IgA, IgM, and IgE) is the most important early screening. Furthermore, the evaluation of specific antibody titers is important. In this regard, serum titers to common vaccinations such as diphtheria, tetanus, and pneumococcus can be used. ELISA is frequently used to detect antibody titers in the serum.

Table 2 Laboratory diagnostic tests for evaluating primary immunodeficiencies.

<i>Laboratory diagnostic tests</i>	
Initial screening	• CBC and differential (platelet volume, absolute lymphocyte count, neutrophil and eosinophil counts) • Immunoglobulin levels (IgG, IgM, IgA) • Lymphocyte subpopulations • Vaccine titers (e.g., diphtheria, tetanus, and pneumococcus) • Complement assessment (e.g., CH50, AH50)
Humoral immunity deficiency	• CBC and differential • Quantitative immunoglobulin analyzes (IgG, IgA, IgM, and IgE) • Specific antibody titers • B cell immunophenotyping (e.g., IgM, IgD, IgG, CD19, CD20, CD27, and CD38) • Genetic mutational analysis (BTK, TACI, BAFFR, ATM, CD79a, CD79b, BLNK, LRRC8, AID, UNG) • KREC
Cellular immunity deficiency	• CBC and differential • T cell immunophenotyping (e.g., CD3, CD4, CD8) • T cell cytotoxicity assay • CD45RA/RO status • DTH tests • Stimulation with mitogens • TREC • Genetic mutational analysis
Combined immunodeficiency	• CBC and differential • Immunophenotyping: T, B, NK cells • Genetic mutational analysis RAG1, RAG2, IL-2 gamma chain, JAK3, CD3, Artemis, Cernunnos, Ligase 4, IL-7 receptor alpha, ADA, PNP, AK2, ZAP70, CD45, DNA-PK, NHEJ1, CD40, CD40L
Innate immunity deficiency	• CBC and differential • Absolute neutrophil count, and morphological analysis of neutrophils • CD11, CD18, CD15c levels: evaluating for LAD • Integrin function • DHR assay • NBT assay • G6PD analysis • NK cytotoxicity and NK markers • Complement assay • Genetic assessment: ELA2, GFI-1, HAX1, GCSFR, MPO

CRP, C-reactive protein; *Ig*, Immunoglobulin; *CH50*, Hemolytic complement (classic pathway); *AH50*, Hemolytic complement (alternative pathway); *BTK*, Bruton's tyrosine kinase; *TACI*, Transmembrane activator and CAML interactor; *BAFFR*, B-cell activating factor receptor; *ATM*, Ataxia telangiectasia mutated; *BLNK*, B-cell linker protein; *LRRC8*, leucine-rich repeat-containing 8; *AID*, Adenosine deaminase; *UNG*, Uracil N-Glycosylase; *KREC*, Kappa-deleting recombination excision circle; *DTH*, Delayed-type hypersensitivity; *TREC*, T-cell replication excision circle; *RAG*, Recombination-activating gene; *JAK3*, Janus kinases; *ADA*, Adenosine deaminase; *PNP*, purine nucleoside phosphorylase; *AK2*, Adenylate Kinase 2; *ZAP70*, Zeta-chain-associated protein kinase 70; *NHEJ1*, non-homologous end joining-1; *CD40L*, CD40-Ligand; *LAD*, Leukocyte adhesion deficiency; *DHR*, Dihydrorhodamine; *NBT*, Nitro blue tetrazolium; *G6PD*, Glucose-6-phosphate dehydrogenase; *NK*, Natural killer cell; *ELA2*, Elastase 2; *GFI-1*, Growth Factor Independent 1; *HAX1*, HCLS1 Associated Protein X-1; *GCSFR*, Granulocyte colony-stimulating factor receptor; *MPO*, Myeloperoxidase.

There are several advanced methods which should be performed on a patient suspected of humoral immune deficiency. Analysis of the expression of B cell markers can be used to assess the B cell phenotype as well as the maturation state of B cells. Important markers used for B cell immunophenotyping include IgM, IgD, IgG, CD19, CD20, CD27, and CD38. IgM and IgD are expressed on mature B cells, but CD27 and CD38 are absent on these cells. CD27 can represent active/memory B cells and CD38 is a marker for antibody-secreting cells and plasma cells. For example, in a patient with common variable immunodeficiency (CVID) decreased levels of memory B cells (CD27⁺, IgM⁻, IgD⁻) are found. Or in patients with X-Linked Agammaglobulinemia (XLA or Bruton), a low or absent circulating B cell population is seen (Aghamohammadi et al., 2005; Moin et al., 2004).

Genetic mutational analysis is another important assessment to detect certain immunodeficiency. This approach can be performed by genome sequencing accompanied with flow cytometry for analyzing the levels of associated protein. For example, a genetic defect in the Bruton's Tyrosine Kinase (BTK) gene is linked to XLA, while a defect in the transmembrane activator and calcium-modulating cyclophilin ligand interactor (TACI) and BAFFR genes is linked to CVID. Or alterations of the ataxia-telangiectasia mutated (ATM) gene are associated with Ataxia-Telangiectasia. By using flow cytometry, intracellular expression of BTK, TACI, BAFFR, and ATM can be identified (Moin et al., 2004; Mohammadi et al., 2009; Zaki-Dizaji et al., 2017; Losi et al., 2005).

Kappa-deleting recombination excision circle (KREC) analysis is another approach to identifying a patient with a humoral immunodeficiency disorder. KRECs are episomal DNA fragments produced during the development of B lymphocytes, importantly when the kappa light-chain genes are being rearranged. PCR can be used to detect KRECs in newborn blood spots, providing a screen for B cell-related deficiencies like XLA and ataxia-telangiectasia (Serana et al., 2013).

Cellular immunity deficiency

In a patient who is suspected of having cellular immunodeficiency, a CBC with differential can be performed to distinguish infants with low/absent absolute lymphocyte counts. Subsequently, flow cytometry and in vitro functional tests can be applied to evaluate the quantity and function of T lymphocytes. Flow cytometry can distinguish between CD3, CD4, CD8, natural killer (NK), and B cell numbers. Immunophenotyping is also important for the diagnosis of severe combined immunodeficiency (SCID). CD3, CD4, and CD8 are T lymphocyte markers that are frequently decreased in defects associated with cellular immunity such as DiGeorge Syndrome. CD45RA is a marker identifying naïve T cells in infants, while CD45RO is considered as a memory T cell marker for identifying maternal T cells. In Omenn syndrome, a variant of SCID, increased frequencies of CD45RO+ T cells are found (Locke et al., 2014; Shahbazi et al., 2019).

Another method for assessing T cell function is the cutaneous delayed-type hypersensitivity (DTH) tests. In these tests, the cellular mediated memory response is detected by intradermal injection of different antigens, including purified protein derivative (PPD), *Candida albicans*, and mumps. The result is assessed 48–72 h later by measuring cutaneous induration. It should be noted that DTH is not recommended for children less than 12 months of age.

The Lymphocyte mitogen assay is another important test for assessing T cell function. In this test, T-cell mitogens are used to stimulate the lymphocytes. The quantification is performed by adding radiolabeled thymidine that is able to be incorporated into the DNA of proliferating cells. Anti-CD3 antibodies, Phytohemagglutinin (PHA), concanavalin (ConA), pokeweed (PWM), and *Escherichia coli* lipopolysaccharide (LPS) are commonly used as mitogens. In this regard, anti-CD3 antibodies, PHA, and ConA are T cell mitogens; LPS is a B cell mitogen; PWM can induce both T cells and B cell responses (Raszka et al., 1996; Ahmed and Blose, 1983).

T-cell replication excision circle (TREC) was recently developed for earlier screening of newborns with severe T cell lymphopenia. Assessing patients using TREC has helped to improve early diagnosis and reduce morbidity and mortality. TREC is referred to the circular, non-replicating pieces of DNA produced during the rearrangement of T cell receptor and they can be considered as a marker for naïve T cells (Somech, 2011).

Genetic investigation of certain genes is considered as another approach to detect cellular immunity defects as well as combined immunodeficiency in which both cellular and humoral immune systems are affected. The genetic assessment can be performed along with the analysis of protein level in the patient's sample.

Quantitative real-time PCR provides the detection of mutations with high sensitivity and specificity. For example, a mutation in the WAS protein (WASp) is associated with WAS. The detection of WASp using Flow cytometry can rapidly screen for this disorder. However, there is a newer method to detect gene alterations known as fluorescent in situ hybridization (FISH). Deletions at 22q11 that occur in DiGeorge Syndrome can be detected by this method (Stewart et al., 2007; Stachon et al., 2007).

Various phenotypes of SCID have been identified by their causative genetic deficiencies, including T-/B+/NK-, T-/B+/NK+, T-/B-/NK+, T-/B-/NK-. Genetic analysis and flow cytometry can detect defected genes. Common gamma chain (CD132) or Janus Kinase 3 (Jak3) deficiency can lead to X-linked SCID (T-/B+/NK- phenotype), the most frequent type of SCID. Deficiency in IL-7 receptor alpha, CD3, CD45, ZAP70, and Coronin-1A are related to T-/B+/NK+ Phenotype of SCID. The T-/B-/NK+ phenotype is associated with gene defects in recombination activating genes-1 (RAG-1) or RAG-2, Artemis, Ligase4 (LIG4), DNA-dependent protein kinase (DNA-PK or PRKDC), and Cernunnos (NHEJ1). In addition, adenosine deaminase (ADA), purine nucleoside phosphorylase (PNP), and adenylate kinase 2 (AK2) deficiencies are associated with the T-/B-/NK- phenotype (Fallah et al., 2018; Locke et al., 2014).

Innate immunity deficiency

There are several disorders related to deficiencies in the innate immune system. Neutrophil-associated disorders are usually characterized by recurrent skin and respiratory tract infections. Other manifestations include delayed umbilical cord separation, poor wound healing, recurrent oral stomatitis, and omphalitis. CBC, absolute neutrophil count, and morphological analysis of neutrophils are considered as initial screening tests (Duchamp et al., 2016; Wong et al., 2013).

Leukocyte Adhesion Deficiency (LAD) is characterized by recurrent bacterial infections, delayed umbilical cord separation, or omphalitis. Three different types of LAD are defined by different gene mutations in molecules involved in leukocyte adhesion. The decreased level of expression of CD11 or CD18 using flow cytometry is considered as a diagnostic test for LAD type 1 (LAD-1). Abnormal expression of CD15c (sialyl lewis X) on neutrophils by flow cytometry is suggestive of LAD-2. In another type of LAD, LAD-3, the function of adhesion molecules is impaired. Therefore, analyzing the function of integrins in leukocytes or platelets is helpful in diagnosing LAD-3 (Etzioni, 2007; Parvaneh et al., 2010).

Chronic granulomatous disease (CGD) is another deficiency related to a defect in the innate immune system components. CGD is characterized by recurrent bacterial and fungal infections due to reduced or absent oxidative burst. CGD can be diagnosed through flow cytometric assessment of the neutrophil oxidative burst by the dihydrorhodamine (DHR) assay. This test is performed by incubation of neutrophils with DHR dye. After that, neutrophils are stimulated by adding phorbol myristate acetate (PMA). PMA induces oxidative burst in neutrophils, resulting in DHR oxidization and fluorescence production. This test is usually abnormal in a patient with CGD. Nitro blue tetrazolium (NBT) is another colorimetric assay previously defined for CGD evaluation. In general, DHR is more sensitive than NBT (Movahedi et al., 2004; Arnold and Heimall, 2017).

Two other disorders also have similar manifestations to CGD, including myeloperoxidase deficiency and glucose-6-phosphate dehydrogenase deficiency (G6PD). Although the DHR assay can be a testing option for diagnosing an MPO-deficient patient,

histochemical staining of MPO is considered as the definitive diagnosis of this disorder. G6PD is also detected by DHR, but the assessment of G6PD activity is also required in patients with recurrent infections or chronic anemia (Frank, 2005; Froissart et al., 2011).

Defects in NK cell numbers and their function result in recurrent infections and the clinical phenotype of hemophagocytic lymphohistiocytosis (HLH), respectively. For screening of suspected patients, immunophenotyping should be performed to verify the existence or absence of NK cells. CD16, CD56, and CD57 markers are usually used for immunophenotyping of NK cells using flow cytometry. On the other hand, evaluation of NK cell function can be performed by a cytotoxicity assay. This test can be done by incubating the peripheral blood mononuclear cells (PBMCs) isolated from suspected patients with labeled target cells. Subsequently, analyzing the markers associated with cell death or apoptosis, such as annexin V by flow cytometry, should be performed. In addition, assessment of CD107a by flow cytometry can be done to investigate the existence of intracellular cytotoxic proteins such as perforin and granzyme. CD107a is a marker involved in cell degranulation. Importantly, the defective expression of CD107a is considered as a diagnostic marker in patients with Griscelli syndrome, Chediak-Higashi, and familial HLH. Cytotoxicity assay can also be done for evaluation of the cytotoxic function of cytotoxic T lymphocytes (CTLs), the cells involved in cellular immunity (Orange, 2013).

Another defect in the innate immune system, complement system deficiency, can be evaluated by analyzing the level or function of the complement system using CH50 and AH50 methods. The hemolytic activity of the classic pathway is measured by CH50 assay, while the function of the alternative pathway is measured by AH50. Therefore, the deficiency in the components of the classic pathway, including C1, C2, and C4 deficiency, leads to a decreased level of CH50 but a normal AH50. On the other hand, a low level of AH50 (with normal CH50) can suggest a defect in the alternative pathway's components such as factor B, factor D, or properdin (Kirschfink and Mollnes, 2003; Frank, 2000; Pettigrew et al., 2009).

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